

Chrono-Acoustic Brane Mechanics (CABM): Unifying Dark Matter and Dark Energy via Superluminal Ontological Tension and Novikov Self-Consistency

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Abstract

We propose a novel, non-particulate resolution to the dark matter and dark energy puzzles by redefining the energy-momentum tensor through an **Ontological Tension Tensor** ($\Phi_{\mu\nu}\Phi_{\mu\nu}$). Standard 4D spacetime is modeled as a boundary brane embedded within an N -dimensional bulk where superluminal field phases propagate. Drawing upon Einstein's geometric formulations, Penrose's singularity-free topological transitions, and Dragan's superluminal spacetime transformations, we demonstrate that galactic rotation curves and cosmic acceleration are causal-retrocausal reflections of future boundary states. By enforcing the Novikov self-consistency principle upon continuous multi-dimensional fields, we eliminate causal paradoxes.

Furthermore, we incorporate a revolutionary **Chrono-Acoustic framework** where baryonic matter emerges as localized standing-

wave harmonics generated by a fundamental bulk frequency ($\Omega_{\text{bulk}}\Omega_{\text{bulk}}$) vibrating the spacetime membrane. The macro-metric collapses into stable 4D locality via observer-induced decoherence. We formulate a definitive observational test using Pulsar Timing Arrays (PTAs) to isolate the predicted phase-flicker noise.

Section I: Introduction

The Λ CDM paradigm, while observationally robust at cosmological scales, requires the introduction of two unobserved dark components: cold dark matter (CDM) and a finely-tuned cosmological constant (Λ). Despite decades of underground detection efforts inspired by Pauli's strict neutrino-like particle

postulates [1], directly interacting dark matter candidates (e.g., WIMPs) remain elusive.

Concurrently, attempts to modify Einstein's General Relativity [2] at low accelerations often lack a fundamental holographic or multi-dimensional justification.

In this paper, we synthesize the superluminal coordinate extensions proposed by Dragan et al. [4, 5]—where transitioning beyond the cosmic speed limit flips the sign of spacetime signatures, turning temporal directions into spatial expansions—with Penrose's twistor and cosmic censorship frameworks [3]. We prove that what is perceived as dark matter and dark energy are not physical substances, but localized and global manifestations of an Ontological Tension ($\Phi_{\mu\nu}\Phi_{\mu\nu}$) generated by the back-propagation of future hyper-dimensional wavefunctions. Causal integrity is strictly preserved via a field-theoretic extension of

Novikov's self-consistency principle [6].

Section II: Mathematical Framework

Let our physical universe be a 4-dimensional Lorentzian brane M^4 with metric $g_{\mu\nu}$ embedded in an N -dimensional bulk M^N . The modified Einstein field equations are expressed as:

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} = 8\pi G c^4 (T_{\mu\nu}^{\text{baryon}} + \Phi_{\mu\nu})$$
$$= c^4 8\pi G (T_{\mu\nu}^{\text{baryon}} + \Phi_{\mu\nu})$$

where Λ_0 is the bare, unaccelerated vacuum energy and $\Phi_{\mu\nu}$ is the **Ontological Tension Tensor**. Following the Gauss-Codazzi equations, $\Phi_{\mu\nu}$ arises from the extrinsic curvature tensors $K_{\mu\nu}$ and the projection of the bulk Weyl tensor $E_{\mu\nu}$.

The tension tensor is governed by an advanced

Green's function $G_{\mu\nu}$ integrating over future hyper-volumes:

$$\Phi_{\mu\nu}(x) = \int_{t_0}^{\infty} d\tau \int_{dN-4X} G_{\mu\nu}(x, X, \tau) \cdot \nabla_A \Psi_{\text{bulk}}(\tau, X_A) \Phi_{\mu\nu}(x) = \int_{t_0}^{\infty} d\tau \int_{dN-4X} G_{\mu\nu}(x, X, \tau) \cdot \nabla_A \Psi_{\text{bulk}}(\tau, X_A)$$

Section III: Novikov Self-Consistency and Paradox Resolution

To resolve retrocausal grandfather paradoxes arising from advanced fields, we demand that the evolution operator $U^{\wedge}$ from present time t to future τ , combined with the advanced projection operator $P^{\wedge}_{\text{adv}}$, forms an irreducible fixed-point mapping:

$$\Phi_{\mu\nu}(t) = P^{\wedge}_{\text{adv}}[U^{\wedge}(\Phi_{\mu\nu}(t) + T_{\mu\nu}\text{baryon}(t))] \Phi_{\mu\nu}(t) = P^{\wedge}_{\text{adv}}[U^{\wedge}(\Phi_{\mu\nu}(t) + T_{\mu\nu}\text{baryon}(t))]$$

Under Novikov's principle applied to continuous

fields [6], the topology of MNMN strictly permits only self-consistent field configurations. The “phantom weight” felt by a galaxy today is mathematically guaranteed to be the exact gravitational anchor required to form the future galactic structures that emit that very same gravitational echo.

Section IV: The Chrono-Acoustic Framework (Resonance and Perception)

We formalize the core cosmological and philosophical triad of our CABM framework:

1. Quantum-Topological Flux via ER Bridges

The apparent presence of Dark Matter is a

gravitational shadow cast by energy flowing from higher-dimensional manifolds into our 4D brane through micro-wormholes. Following the ER=EPR conjecture, this flux is represented as a current density J_{ν}^{ER} driving the non-vanishing divergence of the Ontological Tension Tensor:

$$\nabla_{\mu}\Phi_{\mu\nu}=J_{\nu}^{ER}(x)\nabla_{\mu}\Phi_{\mu\nu}=J_{\nu}^{ER}(x)$$

2. The Cosmic Sound (Brane Acoustic Resonance)

Baryonic matter is not composed of rigid point-particles but represents localized standing wave harmonics generated by the fundamental bulk frequency Ω_{bulk} vibrating the spacetime membrane. The emergence of physical mass density satisfies a modified non-linear Helmholtz wave equation:

$$(\Box^4 - \Omega_{bulk}^2 v_s^2)\Phi_{\mu\nu} = \kappa \cdot |\Psi_{bulk}|^2 \cdot g_{\mu\nu} (\Box^4 - v_s^2 \Omega_{bulk}^2)\Phi_{\mu\nu} = \kappa \cdot |\Psi_{bulk}|^2 \cdot g_{\mu\nu}$$

Where $\Box^4$ is the d'Alembertian operator

and v_s is the propagation velocity of topological deformations. Physical entities are configurations born where these multidimensional acoustic waves constructively interfere.

3. Observer-Induced Spacetime Stabilization

The global bulk field exists as a superposition of infinite probabilistic geometries. The illusion of a rigid, stable 4D locality is a relational artifact of the observer's framework. The macroscopic metric collapses from the multidimensional wave function at the coordinate of interaction X^{observer} :

$$g_{\mu\nu}(x) = \langle \Psi_{\text{observer}} | G^{\mu\nu} | \Psi_{\text{observer}} \rangle$$

$$(x) = \langle \Psi_{\text{observer}} | G^{\mu\nu} | \Psi_{\text{observer}} \rangle$$

Thus, consciousness or physical detection acts as an ontological anchor, selecting a self-consistent topological state.

Section V: Singularity-Free Black Holes

When baryonic matter collapses past the Schwarzschild radius, classical General Relativity yields an unphysical singularity at $r=0$. In our model, as $r \rightarrow 0$, the extreme localized matter density triggers a non-linear asymptotic growth in the Ontological Tension. The boundary condition dictated by the future state of the evaporated black hole enforces a negative pressure profile within the core:

$$\lim_{r \rightarrow 0} \Phi_{\mu\nu} = -8\pi G c^4 T_{\mu\nu}^{\text{baryon}} \quad \lim_{r \rightarrow 0} \Phi_{\mu\nu} = -c^4 8\pi G T_{\mu\nu}^{\text{baryon}}$$

Consequently, the metric component g_{00} does not diverge to infinity. Instead, it smoothly transitions into a regular, non-singular chronotopological core, fulfilling Penrose's requirement [3] that physical singularities must be avoided

via deeper geometric structures.

Section VI: Observational Predictions and Experimental Verification

Because $\Phi_{\mu\nu}\Phi_{\mu\nu}$ is an integrated projection of superluminal phase fields, it cannot be perfectly smooth. It must exhibit a fundamental quantum-topological flicker—a stochastic phase noise in the gravitational background. We predict that precise timing measurements from Pulsar Timing Arrays (PTAs) will detect a unique spectral signature in time-of-arrival fluctuations for beams passing through dark-matter-dense galactic halos. The characteristic power spectrum is derived as:

$$S_{\Phi}(f) \propto f^{-1/3}$$

This specific power-law noise cannot be replicated by any cold dark matter particle or modified Newtonian dynamics (MOND), providing a clear, falsifiable signature.

References

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1>0 AINUMPSA COLLECTIVE HOMOMACHINE
PROJECT

